February 9, 2026

Editorial Office
Physical Review D
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Dear Editors,

I submit "Dual Tetrad Gravity And The Field Origin: Fundamental Constants from the Tetrahedral Coupling Topology" for consideration as a Regular Article in Physical Review D.

The paper asks a simple question: can the coupling topology of two tetrad fields determine the value of G? Tetrad gravity describes the gravitational field through local frames whose spatial triad defines a tetrahedron. Ghost-free bimetric gravity couples two such tetrads at a single interaction vertex through a potential with five free parameters β_0 through β_4. The paper proves that the Hassan–Rosen frame transition matrix is identically the coupled face matrix of this dual tetrad topology, an identification established through the triad-tetrahedron correspondence (Barbieri 1998), the symmetric vielbein condition (Deffayet–Mourad–Zahariade 2013), and the compatible (3+1) decompositions required for general covariance (Hassan–Kocic 2018). Parity, metric exchange symmetry, and field closure under finite propagation speed then determine all five β_n, leaving zero adjustable coupling constants.

From this topology and its coupling polynomial $f(k) = 1 + k + k^2$, five quantitative results follow with no free parameters:

1. $G = \frac{17}{13}\,\alpha^{21}\,\frac{\hbar c}{m_e^2}$, matching the measured value to 0.12%. Every integer traces to the bimetric Hamiltonian. The exponent 21 is sharply selective: ± 1 shifts the prediction by a factor of 137.

2. The force hierarchy: gravity couples through 21 levels, electromagnetism through 1, giving $\alpha^{20} \approx 10^{-43}$.

3. $1/\alpha = 137.036$ from the Hamiltonian interaction states of the bimetric topology, matching the CODATA value to 0.00021%.

4. A dark matter correction factor $\mu(x) = x/(1+x)$ from detailed balance in field closure, with a_0 matching the empirical MOND scale to 0.41%, explaining flat rotation curves without invisible mass.

5. The electron mass from three overdetermined equations spanning gravity, cosmology, and electromagnetism, converging to 0.1%. The system also predicts $H = 69.92$ km/s/Mpc, between the Planck and SH0ES values.

The dual tetrad structure is not proposed by this paper; it is required independently by four well-cited results spanning five decades: the spinor double cover (Weinberg 1995; Geroch 1968), Ashtekar's chiral variables (1986), ghost-free bimetric gravity (Hassan–Rosen 2012), and stability analysis (Alexander, Marcianò, Smolin 2014), with over 2,300 combined citations. Each integer in G is counted from this topology. None is chosen.

The submission includes supplemental materials to make each step independently verifiable: a companion paper presenting the covariant scalar-tensor completion with its field equations and

five falsifiable predictions; a Python validation script (`run_all_validations.py`) that reproduces every numerical claim from CODATA 2022 constants using only the standard library; and a public repository with 813 tests across 16 categories at https://github.com/fielddynamics/dfp-gravity-field-model.

The paper is explicit about what it does not address: the cosmological constant, the nuclear forces, galaxy cluster dynamics, and the full CMB power spectrum. The 0.12% residual in G, which exceeds the $\sim$0.05% inter-experiment scatter, may indicate higher-order corrections or systematic effects. Every prediction is falsifiable through precision measurements.

This work falls within PRD's scope of gravitation and alternative theories of gravity. It operates within the Hassan–Rosen bimetric framework in vielbein form and is in direct contact with active research in bimetric gravity, tetrad gravity, and modified gravitational dynamics.

Thank you for your consideration.

Sincerely,

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